Assignment 1

Variational Free Energy and Importance Sampling

Overview

Given a reference measure Q_0 over trajectories and a cost functional $C(X)$, the optimal change of measure Q^* minimizes the free energy

$$\mathcal{F}(Q) \;=\; \mathbb{E}_Q\big[C(X)\big] \;+\; \alpha\,\mathrm{KL}(Q \,\|\, Q_0),$$

where $\alpha > 0$ is the temperature parameter. The minimizer satisfies $Q^*(X) \propto Q_0(X)\,\exp\!\big(-C(X)/\alpha\big)$, and the optimal free energy has the dual form

$$\mathcal{F}(Q^*) \;=\; -\alpha \log \mathbb{E}_{Q_0}\!\big[\exp\!\big(-C(X)/\alpha\big)\big]. \tag{6.43}$$

Define the *desirability score* of a sample $X_i \sim Q_0$ as

$$r(i) \;=\; \exp\!\big(-C(X_i)\,/\,\alpha\big). \tag{6.45}$$

The free energy can then be estimated from N i.i.d. samples as $\hat{\mathcal{F}} = -\alpha \log\!\big(\frac{1}{N}\sum_i r(i)\big)$. Sampling from Q^* itself requires reweighting the reference samples by their desirability scores (Algorithm 10).

In this assignment you will implement this method and study its behavior as a function of temperature and sample size.

Implementation

All code is in `variational/sampling.py`. Replace each `raise NotImplementedError` with your implementation.

1. **Desirability scores** (Eq. 6.45)

Implement `desirability_scores(C_values, alpha, xp)`.

Given N cost values and temperature α, compute the unnormalized importance weights $r(i)$. Use min-subtraction for numerical stability.

2. **Free energy estimate** (Eq. 6.43)

Implement `free_energy_estimate(C_values, alpha, xp)`.

Estimate $\mathcal{F}(Q^*)$ from the sample average of desirability scores. Use `desirability_scores()` and account for the min-subtraction offset.

3. Inverse CDF resampling (Algorithm 10)

Implement `inverse_cdf_resample(samples, rewards, n_resample, rng, xp)`.

Given reference samples $\{X_i\}$ and their desirability scores $\{r(i)\}$, draw new samples distributed according to Q^* using inverse CDF resampling.

Experiments

Boltzmann sampling on a 2D landscape

The reference measure R is uniform on $[0,1]^2$, and the cost is a bowl with ripple structure: $C(x,y) = 8\big((x-0.5)^2+(y-0.5)^2\big)+0.8\sin(3\pi x)\sin(3\pi y)+0.5$. The optimal Q^* concentrates near the center where cost is lowest.

```
python examples/variational/boltzmann_sampling.py
python examples/variational/boltzmann_sampling.py --alpha 0.1
python examples/variational/boltzmann_sampling.py --alpha 10.0
```

Temperature sweep on a 1D double-well

The cost is $C(x) = (x^2-1)^2$, a symmetric double-well with minima at $x = \pm 1$ and a barrier at $x = 0$. The reference R is uniform on $[-2,2]$. Five values of α show how the Boltzmann distribution transitions from bimodal to nearly uniform.

```
python examples/variational/alpha_sweep.py
```

1. Run `boltzmann_sampling.py` with $\alpha \in \{0.1,\ 1.0,\ 10.0\}$. How does the free energy convergence rate (in N) depend on α? Relate your observation to the effective sample size $N_{\text{eff}} = \big(\sum_i w_i^2\big)^{-1}$, where $w_i = r(i)/\sum_j r(j)$.

2. For $\alpha = 0.1$ and $N = 1000$, what fraction of the weights w_i are numerically nonzero? What does this imply about the variance of the free energy estimate at low temperature?

3. Run `alpha_sweep.py` and examine the resampled histograms across the five temperature values. At what α does Q^* become effectively deterministic (concentrating on a single mode)?

Optional: GPU acceleration

Run the built-in benchmark and compare CPU vs GPU wall-clock time.

```bash
python examples/variational/boltzmann_sampling.py --benchmark
python examples/variational/boltzmann_sampling.py --benchmark --gpu
```

At what sample size does the GPU overhead become worthwhile?

Deliverables

- Completed `variational/sampling.py`.
- Convergence plots for three values of α.
- Written responses to experiment questions 1–3.